

Homework 23: Applications of Determinants

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BCS A, E, F, G

The Adjoint of a Matrix.

Recalled from lecture discussion and *homework 20* that the cofactor C_{ij} of a square matrix A is $(-1)^{i+j}$ times the determinant of the matrix obtained by deleting the i th row and j th column of A . The matrix of cofactors of A has the form

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}.$$

The transpose of this matrix is the adjoint of A and is denoted $\text{adj}(A)$. That is,

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}^T = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}.$$

Problem 1. Finding The Adjoint of a Square Matrix.

Find the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$$

Theorem 1. The Inverse of a Matrix Using Its Adjoint.

If A is an $n \times n$ invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

Problem 2. Use the adjoint of A to find A^{-1} , where

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$$

Theorem 2. Cramer's Rule.

If a system of n linear equations in n variables have a coefficient matrix A with a nonzero determinant $|A|$, then the solution of the system is

$$x_1 = \frac{\det(A_1)}{\det(A)}, \quad x_2 = \frac{\det(A_2)}{\det(A)}, \quad \dots \quad x_n = \frac{\det(A_n)}{\det(A)},$$

where

$$A_i \equiv A_i(\mathbf{b}) = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_{i-1} \quad \mathbf{b} \quad \mathbf{a}_{i+1} \quad \cdots \quad \mathbf{a}_{n-1} \quad \mathbf{a}_n].$$

That is, i th column of coefficient A being replaced by $A_i(\mathbf{b})$.

Problem 3. Cramer's Rule.

Use Cramer's Rule to solve the system of linear equations for x .

$$-x + 2y - 3z = 1, \quad 2x + z = 0, \quad 3x - 4y + 4z = 2.$$

Applications in Analytic Geometry – Area, Volume, Equations of Lines and Planes

I. Area of a Triangle in the xy –Plane. The area of a triangle with vertices (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is

$$Area = \pm \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix},$$

where the sign ($\pm$) is chosen to give a positive volume.

Problem 4. Finding the Area of a Triangle.

Find the area of the triangle whose vertices are $(1,1)$, $(2,2)$, and $(4,3)$.

If three points in the xy –plane lie on the same line, then the determinant in the formula for the area of a triangle is zero, as generalized below.

Test for Collinear Points in the xy –Plane. Three points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are collinear if and only if

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

II. Two-Point Form of an Equation of a Line. An equation of the line passing through the distinct points (x_1, y_1) and (x_2, y_2) is

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0.$$

Problem 5. Finding an Equation of the Line Passing Through Two Points.

Find an equation of the line passing through the points $(2,4)$ and $(-1,3)$.

III. Volume of a Tetrahedron. The volume of a tetrahedron with vertices (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , and (x_4, y_4, z_4) is

$$V = \pm \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix},$$

where the sign ($\pm$) is chosen to give a positive volume.

Problem 6. Finding the Volume of a Tetrahedron.

Find the volume of the tetrahedron with vertices $(4,0,0)$, $(0,4,1)$, $(3,5,2)$, and $(2,2,5)$.

If four points in three-dimensional space lie in the same plane, then the determinant in the formula for the volume of a tetrahedron is zero. So, you have the test shown below.

Test for Coplanar Points in Space. Four points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , and (x_4, y_4, z_4) are coplanar if and only if

$$\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0.$$

IV. Three-Point Form of an Equation of a Plane. An equation of the plane passing through the distinct points (x_1, y_1, z_1) , (x_2, y_2, z_2) , and (x_3, y_3, z_3) is

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0.$$

Problem 7. Finding an Equation of the Plane Passing Through Three Points.

Find an equation of the plane passing through the points $(0,1,0)$, $(-1,3,2)$, and $(-2,0,1)$.

Submission Date: October 20, 2024 (Sunday – 11:59PM)

Good Luck